Who Pays for Global Decarbonization? Fossil Rents and the Terms of Trade in a Small Fossil-Fuel Exporter*

Oliver Loertscher

Signal49 Research

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Abstract

How are welfare losses distributed across households in a small fossil-fuel exporting economy during a global transition away from fossil fuels and what channels determine the losses? I build a multi-sector small open economy model with finitely lived households with idiosyncratic earnings risk, and endogenous terms of trade. I calibrate the model to Canadian data and find that aggregate losses across households amount to a 2 percent loss in consumption. The lifetime loss in consumption due to a fall in the world price for fossil exports explains between roughly 75 to 85 percent of the aggregate loss while the domestic carbon tax with a per-capita rebate explains approximately 15 percent. Young households and future generations lose the most. The depreciation in the real exchange rate that is required to replace fossil exports lowers wages, costing young workers while raising the return on savings of older households. A model with a fixed real exchange rate understates the loss of future cohorts by as much as half. Compensating households alive at the start of the transition would cost 28 percent of GDP.

Keywords: carbon pricing; fossil-fuel exporters; terms of trade; overlapping generations; incidence; intergenerational transfers.
JEL codes: E00, F41, H23, Q43, Q52.

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1 Introduction

Global decarbonization presents an adjustment problem for countries that export fossil fuels. The literature on who bears the burden of carbon pricing focuses on the domestic carbon price paid by households and firms (Grainger and Kolstad, 2010; Fried et al., 2018). For small fossil fuel exporters a global decline in demand for fossil fuels also lowers the world price of fossil fuels, affecting a key industry in the economy. The loss of extraction rents and export income is transmitted to households through wages, profits, investment, the terms of trade and the value of foreign assets. The distributional consequences for small fossil fuel exporters cannot be understood solely from the domestic carbon charge alone.

This paper asks who, inside a fossil-exporting economy, bears the cost of global decarbonization, and how are these losses transmitted to households across the age and income distribution? To answer the question, I build a multi-sector small open economy model with overlapping generations, calibrated to Canada in 2022. Households face idiosyncratic earnings risk, buy energy and consumption goods, and make savings choices. Households provide labour and capital to non-energy, clean energy and fossil fuel producers. Fossil fuel is also exported. The final consumption and investment goods is an Armington composite of home goods (produced with non-energy and energy inputs) and foreign wholesale goods. Home goods and imports are imperfect substitutes, so the terms of trade adjust endogenously. Along an announced transition the domestic carbon tax rises while the world fossil price falls. Welfare is measured as the consumption equivalent variation (CEV) along the transition relative to the initial steady state.[1]

Households lose about 2 percent of consumption on average. The fall in the world fossil price accounts for 1.54 percentage points (p.p.) of this loss and the domestic carbon tax with a per-capita rebate accounts for 0.30 p.p. Young households and future generations lose due to both weaker lifetime labour incomes and the fall in income derived from fossil rents. To contextualize these numbers, compensating every household alive at the start of the transition would cost 28 percent of the initial period GDP as a lump sum, more than half of it owed to households aged 20–34.

This paper contributes to three literatures. The first literature concerns falling global demand for fossil fuels. Several papers find that falling fossil demand strands assets and lowers the income of exporting regions (Mercure et al., 2018; van der Ploeg and Rezai, 2020). Arkolakis and Walsh (2023) highlight the welfare losses fossil fuel exporters face in a world moving towards clean energy due to the loss of export revenues. Charnavoki and Dolado (2014) show that global commodity-price shocks are a major driver of Canadian economic fluctuations. This paper expands on this literature by studying the channels that transmit the change in the global price for fossil fuels across the economy and quantifying the distributional impacts for households by age and income. I show that the price shock

[1]The model features no benefits from emissions reduction. The economy takes the climate path as given, and changes in welfare reflect the economic costs of global abatement within the small open economy.

accounts for between 75 to 85 percent of the loss in lifetime consumption of current households. The result holds for every ordering of the model mechanisms and for every assignment of the ownership of rents that I consider. Foreign ownership of a third of the rents lowers the domestic cost of the external shock from -1.65 to -1.25 points but leaves it more than five times the cost of the carbon tax.

The second contribution is to show that the terms of trade decide how the external loss is shared across generations. The literature on resource exporters studies windfalls and their reversal (van der Ploeg, 2011), how resource revenues are managed (van der Ploeg and Venables, 2011) and the current account of exporters of exhaustible resources (Bems and de Carvalho Filho, 2011). The approach has mostly focused on representative-agent settings. Life-cycle models of carbon taxation, in turn, typically assume a single tradable good, where the real exchange rate is fixed. I show that replacing fossil export income requires a depreciation in the real exchange rate of about 15 percent in the long run. The depreciation affects the youngest households by lowering the wage, at a cost of about 1.5 p.p. Concurrently, the domestic interest rate rises above the world interest rate, raising the welfare of retired households by about 0.3 p.p. For future cohorts, the endogenous terms of trade raise the loss from -4.3 to -8.7 percent, a factor of 1.6 to 2.3 across the trade elasticities I consider. A model with a fixed real exchange rate therefore understates the burden of global decarbonization on future generations by about half.

The third contribution concerns the literature on carbon taxation and revenue. Existing research documents the regressive impacts of carbon taxes and the role of income sources (Grainger and Kolstad, 2010; Rausch et al., 2011), the distribution across current and future generations in life-cycle models (Fried et al., 2018; Kotlikoff et al., 2021), and the unequal response to carbon-pricing shocks (Känzig, 2023). Studies of revenue recycling quantify the trade-offs of rebates and tax cuts (Goulder et al., 2019; Douenne, 2020). I contribute to this literature by quantifying the welfare changes from the household and industrial carbon taxes separately. The household charge is regressive before recycling, but a per-capita rebate more than offsets it, leaving net transfers of $+2.56$ percent of spending in the bottom quintile and $+0.88$ in the top in 2050. The net transfer is progressive regardless of how households are ranked. How the revenue is used trades off current against future households: the per-capita rebate is the most progressive use, and an income-tax cut is the most beneficial to future cohorts, raising their welfare by 1.03 p.p. at the expense of the 2022 population.

The rest of the paper is organized as follows. Section 2 presents the model and Section 3 the calibration. Section 4 studies the fossil-price and terms-of-trade channel, its robustness and the cost of compensation. Section 5 turns to household energy and the design of carbon pricing, and Section 6 concludes. Appendix A collects the equilibrium conditions, Appendix B presents robustness exercises.

2 A small open economy with household energy and endogenous terms of trade

The setting is a small open economy. Households are finitely lived and face idiosyncratic income risk. The production side of the economy is a multi-sector structure comprised of non-energy, clean energy, and fossil fuel production. The final consumption and investment good is an Armington aggregate combining a home good (produced with non-energy and energy inputs) and a foreign good. I describe the household block, production and the external fossil-price shock, the trade structure, the real interest rate and the external balance, and the government. Time is discrete and annual starting with $t = 0$, and all real prices are in units of the composite consumption good defined in Section 2.3. Appendix A lists the equilibrium conditions.

2.1 Households

Demographics and income risk. Households enter the economy at age 20 ($j = 0$) and live for at most 80 periods to a maximum age of 100. With probability s_j a household survives from age j to $j + 1$. They work until retirement at age 64 ($j < j^{ret} = 44$) and then receive a pension pre_t. A worker of age j with idiosyncratic productivity z supplying hours h earns $w\varepsilon_j z h$ (with time t subscripts suppressed for notational convenience), where ε_j is a deterministic age-efficiency profile and $\log z$ follows an AR(1) process with persistence ρ_z and variance σ_z^2. Households save in a one-period asset a' and receive three per-capita transfers: accidental bequests beq_t, the rebate of carbon revenue reb_t, and the fossil rents and firm profits π_t^{hh}. Taxable income y is taxed with the Heathcote et al. (2017) function $T(y) = y - \lambda_t y^{1-\tau_p}$ and consumption is taxed at the rate $\tau_{c,t}$.

Household energy. Households consume a non-energy composite c and energy services e. Energy services are a CES aggregate of clean energy e_C and fossil fuel e_F,

$$e = \left[\omega_h^{1/\eta_h} e_C^{(\eta_h-1)/\eta_h} + (1-\omega_h)^{1/\eta_h} e_F^{(\eta_h-1)/\eta_h}\right]^{\eta_h/(\eta_h-1)}, \tag{1}$$

and enter a Stone–Geary CES composite with subsistence level $\bar{e}$,

$$C = \left[(1-\kappa)^{1/\zeta} c^{(\zeta-1)/\zeta} + \kappa^{1/\zeta}(e-\bar{e})^{(\zeta-1)/\zeta}\right]^{\zeta/(\zeta-1)}, \qquad \zeta < 1. \tag{2}$$

Household utility per period is given by

$$u(C,h) = \frac{C^{1-\sigma}}{1-\sigma} - \xi \frac{h^{1+1/\varphi}}{1+1/\varphi}. \tag{3}$$

With $\bar{e} > 0$ the energy expenditure share falls with total spending, so energy is a necessity; with $\zeta < 1$ energy and other consumption are complements.

Households pay a per-unit consumer charge $\tau^h_{F,t}$ on fossil fuel, set independently of the industrial charge $\tau_{F,t}$ paid by firms. Given the clean-energy price $p_{C,t}$ and the fossil price $p_{F,t}$, the price of energy services and the price index of C are

$$p_{e,t} = \left[\omega_h p_{C,t}^{1-\eta_h} + (1-\omega_h)\left(p_{F,t} + \tau^h_{F,t}\right)^{1-\eta_h}\right]^{\frac{1}{1-\eta_h}}, \qquad P_{C,t} = \left[(1-\kappa) + \kappa\, p_{e,t}^{1-\zeta}\right]^{\frac{1}{1-\zeta}}. \quad (4)$$

Demands follow from the CES nests: $c = (1-\kappa)P_C^{\zeta} C$, $e - \bar{e} = \kappa(p_e/P_C)^{-\zeta} C$, $e_C = \omega_h (p_C/p_e)^{-\eta_h} e$ and $e_F = (1-\omega_h)\left((p_F + \tau^h_F)/p_e\right)^{-\eta_h} e$. Total spending can be written as $(1+\tau_{c,t})(c + p_{e,t}e) = \tilde{P}_t C + m_t$, where $\tilde{P}_t = (1+\tau_{c,t})P_{C,t}$ is the effective price of the composite and $m_t = (1+\tau_{c,t})p_{e,t}\bar{e}$ is committed spending on subsistence energy.

Household problem. Households face the budget constraint

$$\tilde{P}_t C_t + m_t + a_{t+1} = a_t + (1+r_t)\, beq_t + reb_t + \lambda_t y_t^{1-\tau_p}, \quad (5)$$

where

$$y_t = \begin{cases} w_t \varepsilon_j z h_t + r_t a_t + \pi^{hh}_t, & j < j^{ret} \\ pre_t + r_t a_t + \pi^{hh}_t, & j \geq j^{ret} \end{cases}.$$

Taxable income includes capital income and the per-capita rents and profits. Accidental bequests are received with one period of interest, distributed uniformly across living households before the start of each period. The real return on assets between periods is given by r_t.

The first-order conditions for hours and savings are

$$\xi \tilde{P}_t h_t^{1/\varphi} C^{\sigma} = \lambda_t (1-\tau_p) w_t \varepsilon_j z\, y_t^{-\tau_p}, \quad (6)$$

$$C_t^{-\sigma} \geq \beta s_j \frac{\tilde{P}_t}{\tilde{P}_{t+1}} \mathbb{E}\left[C_{t+1}^{-\sigma} R^{eff}_{t+1}\right], \qquad R^{eff}_{t+1} = 1 + r_{t+1}\lambda_{t+1}(1-\tau_p) y_{t+1}^{-\tau_p}, \quad (7)$$

Bequests, pensions and transfers. The assets of households who die at the end of period t form a pool that is paid uniformly to every household alive in $t+1$: $beq_{t+1} = pool_t/\text{POP}$, received as $(1+r_{t+1})beq_{t+1}$ in (5). Newborns enter with zero assets and draw z then receive their bequest. Pensions replace a fraction ss of aggregate labour earnings, and pre_t is the bill divided by the mass of retirees. Fossil rents $\pi_{F,t}$ and the profits of the non-energy and clean sectors ($\pi_{NE,t}$, $\pi_{C,t}$); zero in a steady state) are distributed per capita, $\pi^{hh}_t = (\pi_{F,t} + \pi_{NE,t} + \pi_{C,t})/\text{POP}$, and taxed as income.

2.2 Production and the external fossil-price shock

Home good and energy bundle. A competitive home good sector combines non-energy inputs x_{NE} and an energy bundle x_E into the home good Y_t,

$$Y_t = \left[\theta_{NE} x_{NE,t}^{\rho_Y} + \theta_E x_{E,t}^{\rho_Y}\right]^{1/\rho_Y}, \qquad x_{E,t} = \left[\theta_C x_{Ce,t}^{\rho_E} + \theta_F x_{Fe,t}^{\rho_E}\right]^{1/\rho_E}, \tag{8}$$

with $\rho_Y = 1 - 1/\eta_Y$ and $\rho_E = 1 - 1/\eta_E$, where x_{Ce} is clean energy and x_{Fe} fossil fuel used by firms. Cost minimization gives the dual price indices

$$p_{y,t} = \left[\theta_{NE}^{\eta_Y} p_{NE,t}^{1-\eta_Y} + \theta_E^{\eta_Y} p_{E,t}^{1-\eta_Y}\right]^{\frac{1}{1-\eta_Y}}, \qquad p_{E,t} = \left[\theta_C^{\eta_E} p_{C,t}^{1-\eta_E} + \theta_F^{\eta_E} \left(p_{F,t} + \tau_{F,t}\right)^{1-\eta_E}\right]^{\frac{1}{1-\eta_E}}, \tag{9}$$

and the cost shares $s_{E,t} = \theta_E^{\eta_Y} (p_{E,t}/p_{y,t})^{1-\eta_Y}$ of energy in gross output and $s_{FE,t} = \theta_F^{\eta_E} \left((p_{F,t} + \tau_{F,t})/p_{E,t}\right)^{1-\eta_E}$ of fossil fuel in the energy bundle. The home good sector earns zero profits.

Sectors. Non-energy goods and clean energy are produced with constant returns to scale, $x_{NE,t} = Z_{NE} K_{NE,t}^{\alpha} L_{NE,t}^{1-\alpha}$ and $x_{C,t} = Z_C K_{C,t}^{\alpha_C} L_{C,t}^{1-\alpha_C}$. Fossil output has decreasing returns to capital and labour and produces according the Leontief production technology

$$Y_{F,t} = \max\left\{Z_F \left(K_{F,t}^{\gamma} L_{F,t}^{1-\gamma}\right)^{\nu_F}, \mu_F x_{FF}\right\}, \qquad \nu_F < 1, \tag{10}$$

where x_{FF} is intermediate use of fossil fuel in fossil production.

The decreasing returns to both capital and labour in the fossil fuel sector stand in for a fixed resource factor, such as reserves of heterogeneous quality, whose return is the fossil rent π_F. This modeling choice implies rising marginal extraction costs and give the price-taking sector a well-defined supply curve.

Due to the Leontief production structure, the fossil fuel sector uses a fraction $1/\mu_F$ of its output as an input, on which it pays the industrial carbon charge, so that its net revenue per unit of output is $\tilde{p}_{F,t} = p_{F,t} - (p_{F,t} + \tau_{F,t})/\mu_F$ and the rent $\pi_{F,t}$ is the net revenue minus factor payments and adjustment costs,

$$\pi_{F,t} = \tilde{p}_{F,t} Y_{F,t} - R_t K_{F,t} - w_t L_{F,t} - w_t(\Phi_{F,t}^K + \Phi_{F,t}^L) \qquad .$$

The use of fossil fuel as a fixed fraction of output reflects the engineering energy requirements of extraction and upgrading.

Capital is rented at $R_t = r_t + \delta$. Capital used in period t is chosen in t and depreciates at the rate δ, so that investment is $I_t = K_{t+1} - (1-\delta)K_t$ with $K_t = \sum_s K_{s,t}$ for $s \in \{NE, C, F\}$.

Adjustment costs. Changing capital or labour within and across sectors is costly. This detail is a parsimonious way to capture the notion that capital may be sector-specific and

slow to install or retire, and workers may face retraining, search and relocation costs when moving between sectors (Artuç et al., 2010; Dix-Carneiro, 2014). Quadratic adjustment costs make the transition gradual rather than an immediate reallocation at the announcement.[2] Sector s pays

$$\Phi^K_{s,t} = \frac{\phi_k}{2}\Big(\frac{K_{s,t}}{K_{s,t-1}} - 1\Big)^2 K_{s,t-1}, \qquad \Phi^L_{s,t} = \frac{\phi_l}{2}\Big(\frac{L_{s,t}}{L_{s,t-1}} - 1\Big)^2 L_{s,t-1} \tag{11}$$

in units of effective labour, paid at the wage. The adjustment costs enter the labour market, $\sum_s (L_{s,t} + \Phi^K_{s,t} + \Phi^L_{s,t}) = \bar{L}_t$, where $\bar{L}_t$ is the effective labour supplied by households, and the sectors' first-order conditions hold at the effective factor prices

$$R_{s,t} = R_t + w_t\phi_k\big(g^K_{s,t} - 1\big) - \frac{w_{t+1}}{1+r_{t+1}}\frac{\phi_k}{2}\Big[\big(g^K_{s,t+1}\big)^2 - 1\Big], \tag{12}$$

$$W_{s,t} = w_t\Big[1 + \phi_l\big(g^L_{s,t} - 1\big)\Big] - \frac{w_{t+1}}{1+r_{t+1}}\frac{\phi_l}{2}\Big[\big(g^L_{s,t+1}\big)^2 - 1\Big], \tag{13}$$

with $g^K_{s,t} = K_{s,t}/K_{s,t-1}$ and $g^L_{s,t} = L_{s,t}/L_{s,t-1}$. Capital–labour ratios satisfy $K_{s,t}/L_{s,t} = \frac{\alpha_s}{1-\alpha_s}W_{s,t}/R_{s,t}$, the unit-cost conditions are $p_{NE,t} = (R_{NE,t}/\alpha)^{\alpha}(W_{NE,t}/(1-\alpha))^{1-\alpha}/Z_{NE}$ and $p_{C,t} = (R_{C,t}/\alpha_C)^{\alpha_C}(W_{C,t}/(1-\alpha_C))^{1-\alpha_C}/Z_C$ in the constant-returns sectors, and $\gamma\nu_F[p_{F,t} - (p_{F,t}+\tau_{F,t})/\mu_F]Y_{F,t}/K_{F,t} = R_{F,t}$ and $(1-\gamma)\nu_F[p_{F,t} - (p_{F,t}+\tau_{F,t})/\mu_F]Y_{F,t}/L_{F,t} = W_{F,t}$ in the fossil sector. In a steady state the effective prices equal R_t and w_t, and the constant-returns sectors earn zero profits.

2.3 Trade and the terms of trade

Consumption of non-energy goods, investment and government purchases are a CES composite of home goods and imports with import weight α_m and elasticity η_m. The composite is the numeraire, so that

$$1 = (1-\alpha_m)p_{H,t}^{1-\eta_m} + \alpha_m Q_t^{1-\eta_m}. \tag{14}$$

Both the relative price of home goods and the real exchange rate are functions of the (non-fossil) terms of trade $S_t = Q_t/p_{H,t}$. Foreign demand for home goods is $X_t = S_t^{\epsilon_x}X^*$, and the demand for imports is $M_t = \alpha_m Q_t^{-\eta_m}\mathrm{Abs}_t$ with absorption $\mathrm{Abs}_t = c_t + I_t + G_t$. Household energy is bought from the domestic energy sectors. The home goods clearing condition,

$$Y_t = (1-\alpha_m)p_{H,t}^{-\eta_m}\mathrm{Abs}_t + X_t, \tag{15}$$

determines the terms of trade. When fossil export income falls, the economy must export more home goods or import less of the foreign good, which requires a depreciation in the

[2]The adjustment costs contribute little to the welfare loss. Section B.2 raises the adjustment cost of fossil capital tenfold to study stranded capital.

exchange rate Q_t. In the perfect-substitutes limit ($\alpha_m \to 0$, $\epsilon_x \to \infty$), $S = p_H = Q = 1$ and net exports of the final good are a residual that balances trade relative to the exports of fossil fuel.

2.4 The real interest rate and the external balance

Given a constant world interest rate r^w, the domestic interest rate is given by

$$1 + r_{t+1} = (1 + r^w)\frac{Q_{t+1}}{Q_t}. \tag{16}$$

The domestic interest rate exceeds the world rate while the real exchange rate depreciates. Since Q_t is an equilibrium price, the domestic interest rate moves only when the real exchange rate moves.

2.5 Government

The government finances expenditures taxes on income, consumption, and domestic fossil fuel use. Government revenue is used to pay pensions, finance government consumption $G_t = g\, p_{W,t} Y_t$ of the composite and rebate carbon tax revenue back to households. It holds one-period debt with constant value b^g, rolled over every year. The scale parameter λ_t of the income-tax function balances the budget in every period,

$$\int \left[y - \lambda_t y^{1-\tau_p}\right] d\mu_t + \tau_{c,t} X_t + \text{carbon}_t = G_t + ss\, \Sigma_t^y + reb_t \text{POP} + r_t b^g, \tag{17}$$

where $X_t = \int (c + p_{e,t} e)\, d\mu_t$ is pre-tax household spending. Net foreign assets are $\text{NFA}_t = A'_t - b^g - K_{t+1}$, where A'_t is end-of-period household wealth including the bequest pool.[3]

Market clearing and carbon revenue. Non-energy goods clear, $x_{NE,t} = x_{NEy,t}$, where $x_{NEy,t} = (1 - s_{E,t}) p_{W,t} Y_t / p_{NE,t}$ is the home good sector's demand. Clean energy is used by firms and households, $x_{C,t} = x_{Ce,t} + E_{C,t}$, with $E_{C,t} = \int e_C\, d\mu_t$. Fossil output is used by the sector itself, by firms and by households, and the remainder is exported:

$$Y_{F,t} = x_{FF,t} + x_{Fe,t} + E_{F,t} + NX_{F,t}/p_{F,t}, \qquad x_{FF,t} = Y_{F,t}/\mu_F, \qquad E_{F,t} = \int e_F\, d\mu_t, \tag{18}$$

where $NX_{F,t}$ is the value of net fossil exports. Carbon revenue is

$$\text{carbon}_t = \tau_{F,t}\left(x_{FF,t} + x_{Fe,t}\right) + \tau^h_{F,t} E_{F,t}, \tag{19}$$

and it is returned to households as an equal per-capita rebate, $reb_t = \text{carbon}_t/\text{POP}$. GDP is the value added of the home good, energy and fossil sectors plus net fossil exports:

[3] Section 5.4 replaces the per-capita rebate by a cut in the consumption tax or in the income tax.

Table 1: Assumed parameters

Parameter	Description	Value	Source
σ	Risk Aversion	2	Standard
δ	Capital depreciation	0.05	Standard
η_E	Elasticity of substitution in Energy production	3	Papageorgiou et al. (2017)
η_Y	Energy and non-energy elasticity in Wholesale production	0.1	Fried et al. (2022)
η_h	electricity–fossil substitution	0.50	Stern (2012); Serletis et al. (2010)
α_m	import weight in the composite	0.26	Bussière et al. (2013)
η_m	home–import substitution	1.00	Galí and Monacelli (2005)
ϵ_x	export demand elasticity	1.50	Backus et al. (1994)

$\text{GDP}_t = p_{H,t}Y_t + p_{C,t}E_{C,t} + p_{F,t}E_{F,t} + NX_{F,t} - \tau_{F,t}(x_{FF,t} + x_{Fe,t})$, where $p_{H,t}$ is the price of the home good.

The external shock. The small economy takes the world fossil price $p^*_{F,t}$ as given. In composite units the domestic fossil price is

$$p_{F,t} = Q_t\, p^*_{F,t}, \tag{20}$$

where Q_t is the real exchange rate. Global decarbonization is modeled as an exogenous decline in $p^*_{F,t}$. It reduces the value of fossil output, extraction rents and net fossil exports even when the domestic carbon price is held fixed.

3 Calibration

The model is calibrated to a 2022 steady state of the Canadian economy. Eight parameters are taken from the literature, seven parameters are externally calibrated, and fourteen parameters are chosen jointly to match fourteen moments. I assume that the world interest rate $r^w = 0.04$ and the depreciation rate on capital $\delta = 0.05$.

I estimate several parameters directly from the data. The production parameters are presented in Table 2. The capital share in non-energy production is estimated using data on labour compensation[4] and value added[5] net of mining and utilities. I take the average across the years 1997-2020. I get a value of $\alpha = 0.4$. Similarly, the capital share in the clean sector is estimated using data on labour compensation[6] and value added[7] in clean electricity production, taking the average across 2012-2024. I compute that $\alpha_C = 0.7$.

The decreasing returns to scale parameters ν_F is set using data on profits and revenues[8] for "Oil and gas extraction and support services". I impute profits by taking the difference

[4]Table 36-10-0489-01, Statistics Canada
[5]Table 36-10-0402-01, Statistics Canada
[6]Table 36-10-0628-01, Statistics Canada
[7]Table 36-10-0627-01, Statistics Canada
[8]Table 33-10-0500-01, Statistics Canada

between "Sales of goods and services" and "Cost of goods sold." I then compute the profit-to-revenue ratio where profit corresponds to the imputed profits and revenue corresponds to "Sales of goods and services." The parameter ν_F is then given by

$$\nu_F = 1 - \frac{\pi_F}{Y_F}$$

where π_F is profits and Y_F is revenues. This returns a value of $\nu_F = 0.73$. The capital share in Fossil sector γ is computed using data on labour compensation[9] and value added[10] in Oil and Gas extraction and Support activities for oil and gas extraction in the province of Alberta in the years 1997 to 2020. Data for value added in Support activities in oil and gas is missing between 1997-2006. I impute the value of these years by computing

$$\kappa = \frac{1}{10} \sum_{t=1997}^{2006} \frac{Y^S_{F,t}}{Y^S_{\mathcal{M},t}}$$

where Y^S_F corresponds to value added in "Support activities for oil and gas extraction", and $Y^S_{\mathcal{M}}$ corresponds to "Support activities for mining, and oil and gas extraction". I then multiply this fraction by the value added in Support activities for mining, and oil and gas extraction between 1997 and 2006. For Alberta this is a good approximation, as Support activities in oil and gas as a fraction of all support activities in mining is fairly constant, accounting for approximately 96% value added between 2007 and 2020. I then construct a series for value added in the fossil sector as

$$\hat{Y}_{F,t} = \begin{cases} \kappa Y^S_{\mathcal{M},t} + Y_{F,t} & if \;\; t \in \{1997, \ldots, 2006\} \\ Y^S_{F,t} + Y_{F,t} & if \;\; t \in \{2007, \ldots, 2020\} \end{cases}$$

where $Y_{F,t}$ corresponds to the value added in Oil and Gas extraction (excluding support activities). Using the value ν_F, I use data on compensation (denoted $w_t L_{F,t}$) and value added (denoted $\hat{Y}_{F,t}$) to compute

$$\gamma = 1 - \frac{1}{24} \sum_{t=1997}^{2020} \frac{w_t L_{F,t}}{\nu_F \hat{Y}_{F,t}}$$

to arrive at $\gamma = 0.66$. Here, $w_t L_{F,t}$ corresponds to the sum of employee compensation.

The Leontief intermediate parameter μ_F is computed from the Canadian Input-Output tables by taking the ratio of value added over intermediates used. In the Fossil sector production function, $\mu_F = Y_F / x_{F,F}$ where Y_F maps to the value added in the fossil sector and $x_{F,F}$ is fossil intermediates used in this sector. The "fossil sector" corresponds to the sum of "Conventional oil and gas extraction", "Non-conventional oil extraction" and "Support activities for oil and gas extraction" in 2013, and "Oil and gas extraction (except

[9] Table 36-10-0489-01, Statistics Canada
[10] Table 36-10-0402-01, Statistics Canada

Table 2: Estimated Production Parameters

	Parameter	Source	Value
α	Capital share, non-energy	StatsCan Tables	0.4
ν_F	DRS parameter	StatsCan Tables	0.73
γ	Capital share, Fossil	StatsCan Tables	0.66
α_c	Capital share, Clean	StatsCan Tables	0.70
μ_F	Intermediate use of Fossil	I/O Table	18.19

oil sands)", "Oil sands extraction" and "Support activities for oil and gas extraction" for the years 2014-2020. I take the average across 2013-2020 and arrive at $\mu_F = 18.19$.

Next, I estimated the consumption tax rates τ_c and the income tax progressivity parameter τ_p. These values are reported in Table 3. The consumption tax rate is computed using the method employed in Mendoza et al. (1994), Krueger and Ludwig (2016). I sum Taxes on products with Taxes on production in the numerator. The numerator is then divided by the sum of household final consumption expenditure and non-profit institutions serving households' final consumption expenditure and General governments final consumption expenditure, net of the numerator. I then take the average for the series from 2012–2024.[11]

Table 3: Estimated Government Parameters

	Parameter	Source	Value
τ_c	Consumption tax rate	StatsCan tables	0.15
τ_p	Income Tax progressivity	StatsCan tables	0.1232

I estimate the income tax progressivity parameter τ_p by regressing log average post-tax income on log average pre-tax income by income decile for the years 1976-2021. I use the share of post-tax income as regression weights. So, if Y_{AT} is after tax income, we get

$$Y_{AT} = \lambda Y^{1-\tau_p}$$

Taking logs, we get

$$\log Y_{AT} = \log \lambda + (1 - \tau_p) \log Y$$

or

$$\log Y_{AT} = \beta_0 + \beta_1 \log Y$$

I run the regression separately for each year, compute $\tau_p = 1 - \beta_1$ for each year, and then take the average across all years.[12]

[11] Data on taxes is taken from Statistics Canada Table 36-10-0221-01. Data on consumption is taken from Statistics Canada Table 36-10-0699-01.

[12] All data is taken from Statistics Canada Table 11-10-0193-01.

Table 4: Jointly calibrated parameters: targets and model fit (2022 steady state)

Target	Data	Model	Parameter	Value
Clean/fossil energy use of firms	0.340	0.340	Z_C	0.221
Net fossil exports / GDP	0.064	0.064	Z_F	0.837
Government consumption / GDP	0.200	0.200	g	0.220
Household wealth / GDP	5.500	5.503	β	0.988
Hours of the working-age population	0.180	0.180	ξ	796.8
Average pension / average income	0.230	0.230	ss	0.110
Carbon revenue / government revenue	0.021	0.021	$\tau_F = \tau_F^h$ (2022)	0.181
ACF(1) of residual log earnings	0.740	0.740	ρ_z	0.769
S.d. of one-year earnings change	0.530	0.530	σ_z^2	0.145
Household energy / household spending	0.050	0.050	κ	0.0296
Engel slope of energy, Q1 vs Q5	0.7025	0.7024	$\bar{e}$	0.00209
Own-price elasticity of energy (1 year)	-0.200	-0.200	ζ	0.240
Electricity share of household energy	0.290	0.290	ω_h	0.254
Government net debt / GDP	0.482	0.482	b^g	8.92

Notes: All targets are matched jointly. The energy value-added target includes energy sold to households. The carbon-revenue target includes the consumer charge; in 2022 the federal fuel charge and the industrial benchmark were both \$50 per tonne, so a single carbon price applies to both bases. Government debt is consolidated general government net debt excluding CPP/QPP assets (Statistics Canada).

Household energy. The weight κ matches an aggregate household energy share of 5 percent of spending.[13] I take the ratio of total expenditure on electricity, natural gas, other fuels, air conditioners, heating equipment, and charging expenses to total expenditure and average it across 2014–2023.[14]

The subsistence level $\bar{e}$ is calibrated to match the Engel slope implied by the ratio between the bottom and top income quintiles' energy consumption. In the data,[15] energy consumption of the bottom income quintile is 0.46 of that of the top quintile. I match the Engel slope implied by the data, $\ln 0.46 / \ln 0.331 = 0.7025$, which summarizes what the pair of ratios says about the curvature of the energy Engel curve. Given the model's spending ratio, this implies an energy ratio of 0.755, a subsistence share of 29.1 percent of aggregate energy services, and energy budget shares that fall from 5.44 percent in the bottom quintile to 4.84 percent in the top.

The elasticity ζ matches a one-year own-price elasticity of household energy demand of -0.20, in line with the short-run meta-analytic estimates of Labandeira et al. (2017); subsistence makes the elasticity of total energy smaller than ζ. The electricity–fossil elasticity $\eta_h = 0.5$ reflects evidence that residential interfuel elasticities are below one (Stern, 2012; Serletis et al., 2010; Jadidzadeh and Serletis, 2016).

[13] Table 11-10-0222-01, Statistics Canada

[14] The years 2018, 2020, and 2022 are excluded because no data was reported in those years.

[15] Table 25-10-0062-01, Statistics Canada

Trade. The import share $\alpha_m = 0.26$ matches the import content of Canadian final domestic demand (Bussière et al., 2013). The home–import elasticity is $\eta_m = 1$ (Galí and Monacelli, 2005) and the export elasticity is $\epsilon_x = 1.5$ (Backus et al., 1994; Imbs and Mejean, 2015); both are varied in Section B.1.

Government debt. Government debt is consolidated general government net debt, 48.2 percent of GDP in 2022.[16]

4 Fossil rents and the terms of trade

In this section I decompose the baseline welfare loss and describe the macroeconomic adjustment and the role of the terms of trade.[17]

4.1 Decomposing the welfare loss

In the baseline the carbon tax rises from 0.18 in 2022 to 1.46 in 2050 while the world fossil price falls along the announced path. Households alive in 2022 lose 2.00 percent of lifetime consumption on average (Table 5). The fossil price mechanism accounts for -1.54 p.p., and -0.30 p.p. is due to the carbon tax. The interaction between the two is -0.07 points.[18]

The decomposition of the contribution of each mechanism depends on the order in which they are switched on. The Shapley value is the average marginal contribution of a model mechanism over every ordering of the three mechanisms (the fossil price, the carbon tax, and the adjustment costs of capital and labour taken together) and is computed from the transitions in which each subset is switched on independently (Table 5). The decomposition attributes -1.62 p.p. for the fossil price, -0.32 for the carbon tax and -0.05 for the adjustment costs. The range across orderings is narrow: the fossil price contribution ranges from -1.70 to -1.54 p.p. and the carbon-tax contribution from -0.36 to -0.29 points. Switching on labour adjustment before capital adjustment changes the two adjustment steps by less than 0.012 points.

The impact of the fossil price mechanism is large for every group. The bottom quintile loses 1.67 p.p. due to the fall in the global fossil price, while the top quintile loses -1.57. The losses are smaller for the old, due to the shorter remaining horizon of per-capita transfers they receive, as well as the gains they derive from the terms of trade adjustment. The carbon-tax mechanism is progressive, with the bottom quintile deriving negligible losses (-0.02 p.p.). In comparison, the top quintile loses 0.56 p.p., about a quarter of their total lifetime losses. For newborn cohorts the fossil price dominates by a wider margin.

[16] Table 10-10-0147-01, Statistics Canada.

[17] Robustness exercises examining the trade elasticities, the path of the world price and the ownership of the rents can be found in Appendix B.

[18] The adjustment of capital and labour contributes -0.10 and $+0.000$ points.

Table 5: Decomposition of the welfare loss of the 2022 population in the baseline (CEV, %)

Mechanism	All	Income Q1	Income Q5	Age 65+	Newborns
Sequential decomposition					
Fossil price	−1.538	−1.665	−1.565	−0.275	−7.935
Carbon tax with rebate	−0.296	−0.018	−0.559	+0.063	−1.226
Interaction	−0.065	+0.014	−0.115	−0.077	+0.416
Capital adjustment	−0.097	−0.259	−0.008	−0.243	+0.046
Labour adjustment	+0.000	+0.008	−0.011	+0.009	−0.001
Total	−1.996	−1.920	−2.259	−0.523	−8.700
Shapley values					
Fossil price	−1.622	−1.784	−1.637	−0.425	–
Carbon tax with rebate	−0.324	−0.010	−0.611	+0.021	–
Adjustment costs	−0.050	−0.127	−0.011	−0.120	–
Range across orderings (min, max)					
Fossil price	−1.701, −1.538	−1.921, −1.652	−1.705, −1.565	−0.569, −0.275	–
Carbon tax with rebate	−0.360, −0.294	−0.020, +0.001	−0.675, −0.554	−0.015, +0.063	–

Notes: Baseline: household energy, Armington trade, per-capita rebate. Sequential steps: the fossil price and the carbon tax are switched on separately without adjustment costs, their interaction is the remainder of the joint transition without adjustment costs, and capital and labour adjustment costs are added. Shapley values average each mechanism's marginal contribution over the six orderings of the three mechanisms, using the eight transitions with each subset switched on; they add to the total. The range across orderings reports the smallest and largest contribution of each shock over the four orderings (price first or tax first, without or with adjustment costs). Newborns: mean CEV of the cohorts born 2023–2122.

The mean CEV of the cohorts born between 2023 and 2122 is −8.70 percent, of which −7.94 p.p. come from the fossil price and −1.23 from the carbon tax.

Endogenous terms of trade Table 6 reports the difference (in p.p.) between the Armington baseline where the real exchange rate is determined endogenously and the same model that pins the real exchange rate at $Q_t = 1$. The effect is attributed to each additional mechanism in the model by repeating the counterfactual transitions individually. Without endogenous terms of trade the mean loss is −1.66 percent, a difference of 0.33 p.p. The change in the fossil price accounts for −1.36 p.p. of the mean loss, and the carbon tax accounts for −0.31 p.p.

The endogenous terms of trade channel amplifies the losses due to the falling fossil price by 0.18 p.p. when excluding the capital and labour adjustment costs. Once adjustment costs are present, the endogenous terms of trade channel further amplifies losses from the fall in the fossil price by 0.12 p.p. When looking at the contribution of the carbon tax in isolation, it is clear the aggregate losses are driven entirely by the fossil price, as both with and without adjustment costs, the carbon tax and the lump sum rebate improve welfare negligibly. The carbon tax has little interaction with the terms of trade because net fossil exports respond by much less than the world price does.

When the real exchange rate cannot move, the loss of fossil export income is absorbed entirely through changes in quantities and the net foreign asset position. With Armington

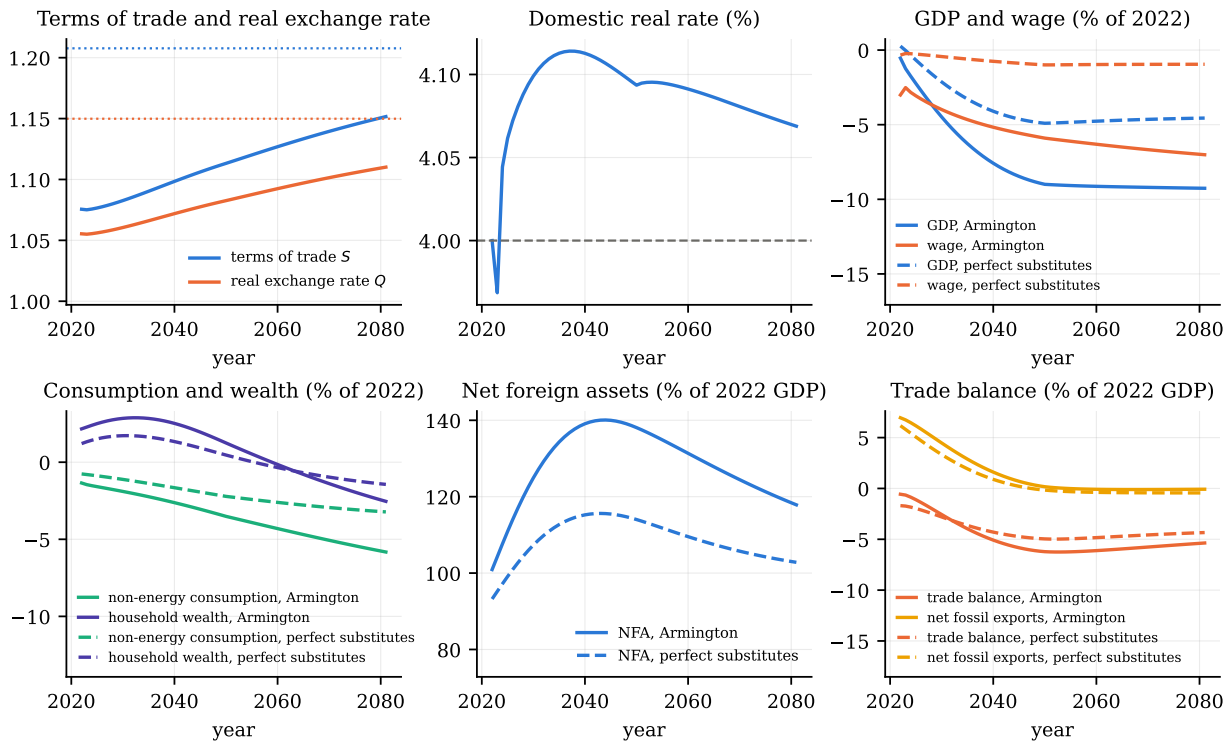


Figure 1: The baseline transition with endogenous terms of trade (annual paths). Terms of trade and real exchange rate (dotted lines: terminal values); domestic real rate (dashed line: world rate); GDP, the wage, non-energy consumption and household wealth under Armington trade (solid) and perfect substitutes (dashed), percent deviation from 2022; net foreign assets and the trade balance.

trade the economy must instead export more of the home consumption goods (or import less of the foreign good). As a result, there must be a depreciation in the real exchange rate causing the terms of trade to rise throughout the transition, and the relative price of the home consumption goods falls. Since the fossil price is expressed in composite units $Q_t p^*_{F,t}$ domestically, the depreciation partly insulates the fossil sector. However, depreciation also reduces the composite-unit value of output and wages (Figure 1).

Relative to a model with a fixed real exchange rate, the endogenous terms of trade adjustment mechanism is an important channel for transmitting the fossil price shock across generations. Adding the Armington structure to the model results in lifetime consumption for the youngest (20–34 year olds) and midcareer (35–49) households to fall an additional 1.50 and 0.53 p.p. respectively. Meanwhile, lifetime consumption for households nearing retirement (50-64) and retired households (65 and older) increases by 0.27 and 0.29 p.p. due to the improved return on savings when the terms of trade adjust. The terms of trade adjustment results in a negligible improvement in consumption due to the carbon tax, between about 0.01 and 0.02 p.p.

The same aggregate depreciation affects cohorts through opposite margins. For the young, the dominant effect is the lower real wage over their remaining working lives. For the retired households, since the domestic real rate rises above the world rate due to the

Table 6: Welfare effect of endogenous terms of trade (CEV, pp) and its attachment to each shock

	All	20–34	35–49	50–64	65+	Q1	Q3	Q5
Armington − perfect substitutes (full transition)	−0.332	−1.497	−0.530	+0.273	+0.292	−0.581	−0.195	−0.298
attached to the fossil price (no adj. costs)	−0.177	−1.569	−0.435	+0.505	+0.617	−0.334	−0.062	−0.196
attached to the carbon tax (no adj. costs)	+0.015	+0.007	+0.020	+0.021	+0.013	+0.008	+0.015	+0.022
attached to the fossil price (full adj. costs)	−0.300	−1.610	−0.485	+0.400	+0.358	−0.616	−0.151	−0.232
attached to the carbon tax (full adj. costs)	+0.012	+0.016	+0.018	+0.009	+0.006	+0.011	+0.009	+0.018
both shocks, no adjustment costs	−0.222	−1.469	−0.493	+0.367	+0.537	−0.318	−0.115	−0.273
	Mean	2030	2050	2070				
Newborn cohorts, CEV (%): perfect substitutes	−4.28	−3.39	−4.15	−4.44				
Newborn cohorts, CEV (%): Armington	−8.70	−5.82	−7.62	−8.80				

Notes: First block: differences in CEV (pp) between the Armington and the perfect-substitutes transitions, cell by cell, for the full transition and for the counterfactual transitions of the decomposition. Last block: mean CEV of the cohorts born 2023–2122 and of the cohorts born in 2030, 2050 and 2070.

exchange rate depreciation, the return on household savings is higher. Retired households have more accumulated assets, and their consumption is more directly impacted by changes in savings income.

Adjustment costs interact with the terms of trade adjustment to slightly increase the losses for all generations when compared to the fixed real exchange rate model. The presence of adjustment costs worsens the difference between the Armington and perfect substitutes models (with both the fossil price and the carbon tax channels active) by about 0.11 p.p. in aggregate. The largest losers are the oldest households, suggesting that adjustment costs interact with the terms of trade channel primarily through wealth rather than wages.

Overall, the poorest households see consumption decline by about 1.92 percent compared to about 1.6 percent for the wealthiest households. Table 5 suggests that the fossil price shock hits the lowest income households the hardest, though the difference from the richest households is about 0.1 p.p. However, the change in consumption directly attributable to the terms of trade shows a hump shaped distribution across quintiles. The poorest households see lifetime consumption decrease by about 0.58 p.p. due to the terms of trade channel, while the richest households are about 0.3 p.p. worse off. Households in the middle of the distribution lose the least (about 0.2 p.p.). The interaction between the adjustment costs and terms of trade appears to tell the story, as the presence of adjustment costs increases the terms of trade impact by about 0.28 p.p. for the first quintile, compared to about 0.08 p.p. for the middle quintile and 0.03 p.p. for the top quintile.

The bottom quintile is composed mainly of young workers with few assets, and retirees with lower lifetime incomes. Young workers incomes are derived primarily from their wages. When the exchange rate is fixed, the value of the home good remains the same in the consumption basket and wages only fall by 0.9 percent in the future steady state. When the exchange rate adjusts endogenously, the value of the home good falls and thus the decline in the wage rate is amplified to about 8.4 percent by the final steady state. Since pensions in the model are a constant fraction of aggregate earnings, the fall in labour income lowers pension payments as well to the detriment of the older households.

The top quintile holds most of the wealth, and the adjustment in the exchange rate compensates some of their labour income losses in the form of higher returns on savings. When the real exchange rate adjusts, the interest on savings temporarily rises above the world interest rate so assets earn a premium along the transition.

4.2 How large is the cost? Compensation

A money-metric measure helps to contextualize the size and the distribution of the losses. For every household alive in 2022 I compute the lump-sum transfer in 2022 that would offset its welfare change and sum over the population. For future cohorts I compute the present value of the same transfers at birth, discounted to 2022 at the world rate.

In the baseline, compensating the 2022 population would cost 28.0 percent of GDP as a lump sum in 2022, of which 14.7 points go to households aged 20–34 and only 1.1 to those over 65. Compensating the cohorts born between 2023 and 2122 would cost a further 40.5 percent of 2022 GDP in present value, or 1.79 percent of 2022 GDP for the cohort born in 2050 alone. With perfect substitutes the two numbers are 22.8 and 22.0 percent. The endogenous terms of trade mechanism add about a fifth to the compensation owed to current households and almost double that owed to future ones.

5 Household energy and the design of carbon pricing

Household energy demand changes the distributional incidence of carbon pricing in two ways. First, it makes energy a direct expenditure for households. Second, it changes how carbon revenue is split between the household and industrial bases. The first effect is regressive before recycling, because energy is a larger share of spending for low-income households. The regressive impacts can be reversed once revenue is returned per capita. This section quantifies both effects, shows how the welfare gradient depends on how households are ranked, compares carbon-pricing instruments at equal abatement and equal revenue, and compares uses of the revenue.

Which environment. The use-side incidence results (Tables 7 and 10) are computed in the perfect-substitutes environment, which isolates the household block and the fiscal rules from the trade structure; in the Armington baseline the use-side incidence is within 0.20 percentage points of spending of the numbers reported here. The *welfare* gradient across households is different: the larger fall of the real wage under Armington trade tilts it against young and low-consumption households. Table 9 reports welfare by quintile.

5.1 Direct incidence and the rebate

In the initial steady state, household energy accounts for 5.44 percent of spending in the bottom income quintile and 4.84 percent in the top quintile. As the carbon charge

Table 7: Use-side incidence by income quintile (% of the quintile's spending; perfect substitutes)

Year		Q1	Q2	Q3	Q4	Q5
2022	Energy budget share	5.44	5.03	4.85	4.96	4.84
	Consumer charge paid	0.59	0.55	0.53	0.54	0.53
	Rebate received	1.27	0.99	0.87	0.94	0.86
	Net transfer	+0.68	+0.44	+0.34	+0.40	+0.33
2050	Energy budget share	7.52	6.92	6.66	6.81	6.64
	Consumer charge paid	4.13	3.80	3.66	3.74	3.65
	Rebate received	6.69	5.21	4.57	4.95	4.53
	Net transfer	+2.56	+1.41	+0.91	+1.21	+0.88

Notes: Perfect substitutes, both charges and the per-capita rebate of total carbon revenue. The rebate includes revenue from the industrial charge. Quintiles of gross income in the year shown; the 2022 values are those of the first year of the transition.

rises along the transition, the real price of energy rises by about 49 percent by 2050. Household fossil use falls by 14 percent and household electricity use rises by 11 percent. The aggregate household energy share rises from 5.0 to 6.9 percent, reaching 7.52 percent in the bottom quintile and 6.64 percent in the top (Figure 2).

Before any rebate, the consumer charge is regressive (Table 7). In 2050 it costs 4.13 percent of spending in the bottom quintile and 3.65 percent in the top. The per-capita rebate, which returns revenue from both the consumer and the industrial charge, is worth 6.69 and 4.53 percent of spending. The net direct transfer is positive in every quintile and progressive, with +2.56 percent of spending for the bottom quintile and +0.88 percent for the top.[19]

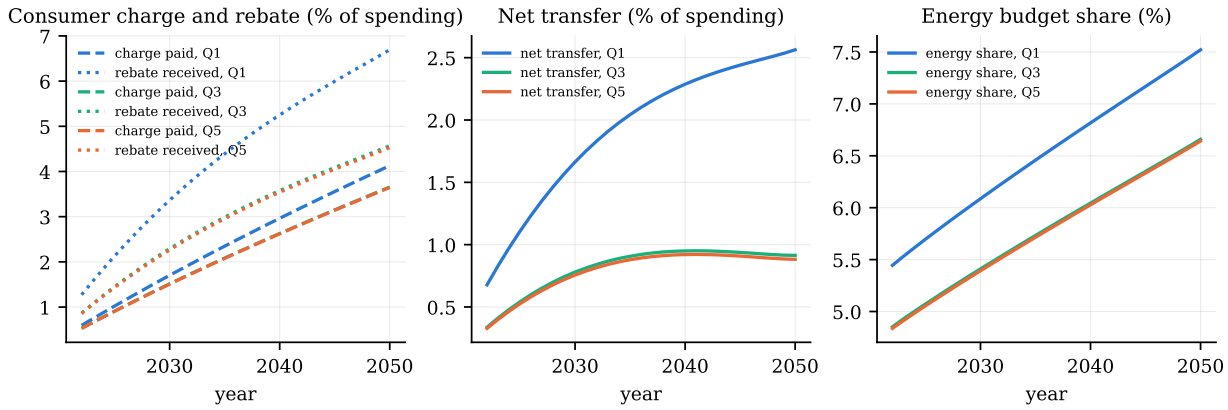


Figure 2: Use-side incidence by quintile of gross income (annual paths, perfect substitutes). Left: consumer carbon charge paid and per-capita rebate received; middle: net transfer; both as a percentage of the quintile's total spending, 2022–2050. Right: energy budget share by quintile, 2022–2050.

[19]The profile of the transfers is not perfectly monotone, since the energy share of the fourth quintile slightly exceeds that of the third.

5.2 Welfare and the ranking of households

The welfare effects remain negative for every quintile. When households are ranked by gross income, lifetime expected consumption is 1.92 percent lower for the bottom quintile and −2.26 percent lower for the top. The difference is not generated by the direct consumer charge, as the carbon-tax component of the decomposition is −0.02 points for the bottom quintile and −0.56 points for the top. Instead, higher income households are more exposed to the general-equilibrium effects of the carbon tax on wages, capital returns and the income-tax scale, even though their direct energy share is lower.

Table 8: Welfare by age and income quintile (CEV, %): Armington baseline and perfect substitutes

Age in 2022	Q1	Q2	Q3	Q4	Q5	All
Armington baseline						
20–34	−4.37	−4.23	−4.20	−3.94	−3.57	−4.06
35–49	−3.31	−3.28	−1.97	−2.59	−2.24	−2.58
50–64	−1.78	−1.49	−0.92	−1.20	−1.11	−1.17
65+	−0.56	−0.58	−0.40	−0.21	−0.07	−0.52
All	−1.92	−1.73	−1.87	−2.20	−2.26	−2.00
Perfect substitutes						
20–34	−2.59	−2.58	−2.60	−2.55	−2.49	−2.56
35–49	−2.16	−2.19	−1.89	−2.05	−1.99	−2.05
50–64	−1.53	−1.46	−1.37	−1.47	−1.46	−1.45
65+	−0.62	−0.91	−0.98	−1.03	−1.05	−0.82
All	−1.34	−1.46	−1.67	−1.88	−1.96	−1.66

Notes: Household energy, per-capita rebate; quintiles of 2022 gross income.

This income gradient includes age effects. Among households aged 20–34 the CEV ranges only from −2.60 to −2.49 percent across quintiles, compared to between −0.62 percent and −1.05 percent among those aged 65 and over in the bottom and top quintiles (Table 8). The bottom gross-income quintile contains many retirees, so ranking by current income conflates income and age. Table 9 therefore reports the CEV with households ranked in three other ways: by composite consumption in the year, by gross income within each of the four age groups, and by the present value of their remaining baseline consumption, a measure of lifetime resources.

Ranking by gross income *within* age groups removes the age component and makes the gradient nearly flat in the perfect-substitutes environment: −1.60 percent for the bottom within-age quintile and −1.68 for the top. In the Armington baseline, the gradient is not quite as flat though it still has a smoothing effect. This is because changes in the interest rate due to the endogenous terms of trade adjustment is transmitted across both the age distribution (primarily to the benefit of older households with more assets) and the income districution (to the benefit of wealthier households).

Ranking by consumption reverses it, and the bottom consumption quintile loses −2.02 percent. The top quintile on the other hand, which contains wealthy older households,

Table 9: Welfare with alternative rankings of households

Ranking of households	Q1	Q2	Q3	Q4	Q5
CEV (%), Armington baseline					
Gross income (benchmark)	−1.92	−1.73	−1.87	−2.20	−2.26
Consumption	−3.26	−2.89	−1.98	−1.22	−0.63
Income within age groups	−2.30	−2.05	−2.10	−1.80	−1.73
Lifetime resources	−0.61	−1.22	−2.72	−3.17	−2.26
CEV (%), perfect substitutes					
Gross income (benchmark)	−1.34	−1.46	−1.67	−1.88	−1.96
Consumption	−2.02	−2.01	−1.70	−1.41	−1.18
Income within age groups	−1.60	−1.65	−1.70	−1.68	−1.68
Lifetime resources	−0.74	−1.32	−2.00	−2.23	−2.03

Notes: Quintiles of the 2022 population by gross income, by composite consumption in 2022, by gross income within each of the four age groups, and by the present value of remaining baseline consumption.

loses −1.18 percent. In the Armington baseline, where the real wage falls more, both rankings tilt against the bottom. When ranked by income within age groups the bottom quintile loses −2.30 percent and the top −1.73, and when ranked by consumption −3.26 against −0.63 percent.

The terms of trade mechanism also affects the ordering of losses, depending on households are ranked. Under the benchmark, when household are ordered by gross income, the economic costs are increasing by income quintile, with the poorest households seeing a consumption decline of 1.34 percent compared to 1.96 percent for the wealthiest households. Introducing the endogenous terms of trade under the Armington specification removes the monotonic ordering across quintiles, with the second and third quintile having smaller consumption declines (1.73 percent and 1.87 percent respectively) compared to the first quintile (1.92 percent).

The choice of household ranking also changes the overall distribution of the impacts of the endogenous terms of trade mechanism. When households are ranked by gross income (or income within age groups), the Armington specification universally amplifies the losses. Ranking households by consumption results in a regressive ordering, increasing the losses for the bottom three quintiles by between 0.28 p.p. and 1.24 p.p., and dampening the losses by between 0.19 p.p. and 0.55 p.p. for the top two quintiles. Ranking instead by lifetime resources results in a more progressive ordering; the bottom two quintiles losses are dampened by between 0.10 p.p. and 0.13 p.p. while the top three quintiles losses are dampened by between 0.23 p.p. and 0.94 p.p.

5.3 Consumer versus industrial carbon pricing

Canada set its consumer fuel charge to zero in April 2025 while retaining carbon pricing for industrial emitters (Department of Finance Canada, 2025). I compare three designs:

Table 10: Carbon-pricing designs: the three designs of Canada's policy debate and the consumer-only design scaled to equal abatement or equal revenue

	Both charges	Industrial only	Consumer only	Consumer only, abatement search (best $\gamma = 4.00$)	Consumer only, equal revenue ($\gamma = 1.39$)
2050, change relative to 2022 (%)					
Domestic fossil use	−34.6	−14.4	−5.5	−16.1	−8.1
households	−14.0	+21.8	−13.8	−32.5	−18.3
firms' inputs	−61.2	−61.3	+5.2	+5.1	+5.2
GDP	−4.91	−4.70	−3.43	−3.58	−3.45
Carbon revenue (% of 2022 GDP)	3.21	0.83	2.67	7.06	3.31
PV of carbon revenue (% of 2022 GDP)	57.1	18.2	46.9	116.6	57.1
Charge in 2050 (composite units)	1.455	1.455	1.455	5.277	1.952
Welfare, CEV (%)					
Mean, 2022 population	−1.663	−1.371	−1.612	−2.406	−1.720
Income Q1	−1.339	−1.375	−1.324	−1.511	−1.343
Income Q5	−1.961	−1.402	−1.849	−3.122	−2.031
Age 20–34	−2.561	−2.256	−2.207	−3.131	−2.335
Age 65+	−0.815	−0.707	−0.960	−1.310	−1.003
Utilitarian	−1.988	−1.698	−1.848	−2.692	−1.963
Newborns 2023–2122, mean	−4.277	−4.203	−3.170	−3.908	−3.265

Notes: Perfect substitutes, per-capita rebate. Domestic fossil use is the sum of the fossil sector's own use, firms' inputs and household use. The first three designs do not hold abatement, revenue or the carbon price constant. The scaled designs multiply the rise of the single-base charge by γ so that 2050 domestic fossil use, or the present value of carbon revenue over the horizon discounted at the world rate, equals that of the combined design (a secant iteration on γ). The abatement search for the consumer-only design was stopped at $\gamma = 4$ without reaching the target; the column reports that last evaluation.

both charges rise as in the baseline; only the industrial charge rises, with the consumer charge zero from 2025; and only the consumer charge rises, with the industrial charge held at its 2022 level.

Under industrial-only pricing, domestic fossil use falls by 14.4 percent by 2050, compared with 34.6 percent when both charges rise (Table 10). Household fossil use rises by 21.8 percent, because the world price falls with no charge to offset it at the pump, so the effective price households pay falls. Carbon revenue in 2050 is only 0.83 percent of 2022 GDP, against 3.21 percent with both charges, so the rebate is much smaller. Consumer-only pricing reduces total domestic fossil use by only 5.5 percent, because firms' fossil inputs rise by about 5 percent as the world price falls, but it raises most of the revenue of the combined design (2.67 against 3.21 percent of 2022 GDP).

For the population alive in 2022, the mean CEV is −1.37 percent under industrial-only pricing, −1.61 percent under consumer-only pricing and −1.66 percent when both charges rise. The distribution differs sharply, and the change in welfare is slightly regressive. Removing the consumer charge reduces the loss of the top quintile from −1.96 to −1.40 percent but leaves the bottom quintile slightly worse off (−1.38 against −1.34 percent), consistent with the fall in its direct net transfer in 2050 from +2.56 to +1.80 percent of spending as the rebate shrinks. Consumer-only pricing is the least costly design for

Table 11: Recycling of carbon revenue in the Armington baseline (CEV, %)

Use of carbon revenue	All	Q1	Q3	Q5	20–34	65+	Newborns
Per-capita rebate (baseline)	−1.996	−1.920	−1.869	−2.259	−4.058	−0.523	−8.70
Consumption-tax cut	−1.930	−2.488	−1.573	−1.726	−4.115	−0.615	−9.28
Income-tax cut (through λ_t)	−2.584	−2.943	−2.323	−2.373	−4.057	−1.614	−7.67

Notes: Armington baseline. Newborns: mean CEV of the cohorts born 2023–2122.

newborn cohorts, whose mean loss is about one point smaller than under the other two designs, reflecting a smaller decline in GDP along the transition.

5.4 Rebates, consumption-tax cut and income-tax cut

The way revenue is recycled matters as much as the base on which it is raised (Table 11). In the Armington baseline, using carbon revenue to cut the consumption tax instead of paying a per-capita transfer raises the mean CEV of the initial population by 0.07 p.p.. The tax cut lowers the CEV of the bottom quintile by 0.57 points and raises that of the top quintile by 0.53 points, acting as a redistribution mechanism from the poor to the wealthy. It also lowers the welfare of future cohorts, by 0.58 points on average.

If the carbon tax revenue is instead used to lower the income tax through λ_t, the mean CEV changes by -0.59 points relative to the rebate. The bottom quintile's welfare falls by -1.02 and that of the top quintile by -0.11 p.p., while that of future cohorts increases by 1.03 p.p.. A cut in the tax on labour and capital income shifts resources from retirees and low-income households to workers and to the cohorts that will work during the transition. The per-capita rebate is the most progressive use of the revenue, while the income-tax cut is the most favourable to future cohorts, at the expense of the initial population and particularly of its bottom quintile.

6 Conclusion

If the world transitions away from fossil fuels, small fossil fuel exporters are impacted by both the domestic carbon price and changes in the terms of trade. Using a quantitative model calibrated to Canada, I show that the decline in the world fossil price accounts for -1.54 of the -2.00 percentage points of consumption-equivalent loss among households, against -0.30 p.p. for the domestic carbon tax with lump sum rebates. The ownership structure of the fossil rents does not alter the dominance of the price channel, but it does alter the ordering of who bears the losses.

Endogenous trade adjustment adds 0.33 points to the loss of current households, particularly impacting households aged 20–34. Compensating the households alive at the start of the transition would cost 28 percent of 2022 GDP.

When energy is a necessity in household consumption choices, the carbon tax paid by

households is regressive, but a per-capita rebate more than offsets it for every quintile regardless of how households are ranked. The use of the revenue trades off current against future households: a lump sum per-capita rebate is the most progressive use of carbon tax revenue, while financing an income-tax cut is preferred by future cohorts.

Assessments of decarbonization in fossil exporters should distinguish the domestic carbon tax from changes in the international price of fossil fuels. Carbon taxes determine direct energy incidence and the revenue available for recycling, and the world fossil price determines the external income shock, the terms-of-trade adjustment and, ultimately, the burden on young and future households. The impact of the global fossil price is magnified via changes in the real exchange rate as the terms of trade adjust in response to the shock.

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A Equilibrium

A.1 Equilibrium

Given exogenous paths $\{p^*_{F,t}, \tau_{F,t}, \tau^h_{F,t}\}$, the world real rate r^w, the initial distribution μ_0 and initial sectoral capital and labour, an equilibrium is a set of household decision rules for c, e_C, e_F, h and a', distributions μ_t, firm choices, prices w_t, $p_{NE,t}$, $p_{C,t}$, $p_{E,t}$, $p_{H,t}$, Q_t and S_t, and government variables λ_t, pre_t, reb_t and G_t such that, in every period $t = 0, \ldots, T$ and with the terminal steady state imposed at $T + 1$:

1. households solve the problem (5)–(7) given prices, transfers and the returns (16), and μ_{t+1} follows from the decision rules, survival, the entry of newborns with zero assets and the transition matrix Π;
2. the wholesale sector's unit cost equals the wholesale price, $p_{y,t} = p_{W,t} = p_{H,t}$, with $p_{y,t}$ and $p_{E,t}$ given by (9); the non-energy, clean and fossil sectors satisfy their first-order conditions at the effective factor prices (12)–(13) with rental rate $R_t = r_t + \delta$ in composite units;
3. UIP (16) holds;
4. markets clear: home goods (15); non-energy goods, $x_{NE,t} = x_{NEy,t}$; clean energy, $x_{C,t} = x_{Ce,t} + E_{C,t}$; fossil fuel (18); labour, $\sum_s (L_{s,t} + \Phi^K_{s,t} + \Phi^L_{s,t}) = \bar{L}_t$; and domestic assets, with net foreign assets equal to household wealth minus government debt minus the capital of firms;
5. the government budget (17) holds with λ_t adjusting, carbon revenue is given by (19), the pension, the rebate and government purchases follow their rules, and bequests, distributed rents and profits are consistent with the distribution and with the sectors' accounts;
6. the balance of payments holds, including the revaluation of foreign assets at the announcement: $\text{NFA}_t = (1 + r^w)\frac{Q_t}{Q_{t-1}}\text{NFA}_{t-1} + p_{H,t}X_t - Q_t M_t + NX_{F,t}$ (minus the rents paid abroad in the variant with foreign ownership).

In the perfect-substitutes structure, $S_t = p_{H,t} = Q_t = 1$, $r_t = r^w$, and net exports of the final good are the residual of the goods market. The counterfactual transitions of the decomposition hold the world price or the carbon charges at their 2022 values and, where indicated, set $\phi_k = \phi_l = 0$; the terminal steady state of each counterfactual is the stationary equilibrium of its own terminal prices.

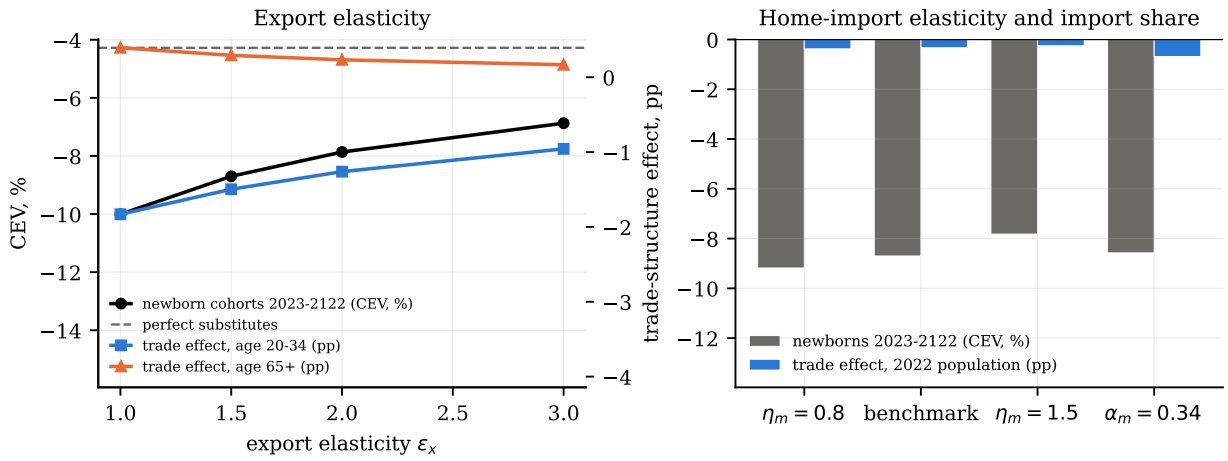


Figure 3: The intergenerational result and the trade elasticities: mean CEV of the cohorts born 2023–2122 and the trade-structure effect on the 2022 population by age, against the export elasticity (left) and for the home–import elasticity and the import share (right).

B Robustness exercises

B.1 Trade elasticities and the import share

The size of the real depreciation, and with it the intergenerational result, is governed by the export elasticity ϵ_x, the home–import elasticity η_m and the import share α_m. I re-solve the terminal steady state and the baseline transition for $\epsilon_x \in \{1, 2, 3\}$, $\eta_m \in \{0.8, 1.5\}$ and $\alpha_m = 0.34$.

A higher export elasticity or a higher home–import elasticity requires a smaller depreciation for the same loss of fossil income, so the trade-structure effect shrinks: with $\epsilon_x = 3$ the terminal terms of trade rise to 1.10 instead of 1.21, the mean effect on the 2022 population is -0.21 instead of -0.33 points, and the loss of newborn cohorts is -6.9 instead of -8.7 percent. A lower export elasticity does the reverse ($\epsilon_x = 1$: a terminal terms of trade of 1.305 and a newborn loss of -10.0 percent). A larger import share makes the same movement of the terms of trade cost more purchasing power: with $\alpha_m = 0.34$ the trade-structure effect is -0.68 points and turns negative even for households over 65. The age pattern is the same in every case. Endogenous terms of trade raise the loss of future cohorts by a factor of 1.6 to 2.3 across the elasticities considered; the export elasticity is the parameter on which the number depends most, and the doubling reported above holds for $\epsilon_x \leq 2$.

B.2 The world fossil-price path

The relative size of the two channels depends on how far and how fast the world price falls relative to the rise of the carbon charge. Table 12 varies the terminal level of the world price (0.30, 0.40, 0.56 and 0.70 of its 2022 level; the lowest is the level implied by the International

Table 12: The world fossil-price path: the two channels for alternative terminal levels, timings and credibility (CEV, %)

	2022 population				Price mechanism		Newborns	
Path	Fossil price	Carbon tax	Residual	Total	20–34	65+	Price	Total
Terminal level 0.30 (IEA net zero)	−2.117	−0.294	−0.083	−2.494	−4.248	−0.878	−8.22	−9.14
Terminal level 0.40	−1.984	−0.294	−0.070	−2.349	−4.084	−0.734	−8.48	−9.28
Terminal level 0.56 (baseline)	−1.647	−0.294	−0.054	−1.996	−3.512	−0.508	−7.91	−8.70
Terminal level 0.70	−1.229	−0.294	−0.040	−1.563	−2.684	−0.327	−6.28	−7.20
Decline completed by 2040	−2.232	−0.294	−0.157	−2.683	−4.388	−0.868	−8.31	−9.11
Decline completed by 2060	−1.272	−0.294	+0.029	−1.537	−2.859	−0.317	−7.54	−8.32
Half of the decline announced in 2022 (to 0.78)	–	–	–	−1.267	−2.653	−0.221	–	−5.93
remainder revealed in 2030 (alive in 2030)	–	–	–	−0.863	−1.608	−0.397	–	–

Notes: Armington baseline. Fossil price: CEV (%) of the price-only transition at full adjustment costs; carbon tax: the tax-only transition, which does not depend on the price path; residual: total minus the two; total: the full transition. The carbon charge keeps its timing in the terminal-level variants and follows the price's timing in the 2040 and 2060 variants. The age columns report the price mechanism for the 2022 population except in the partial-credibility rows, which report the total by age. The partial-credibility rows report the CEV of the 2022 population under the half-announced decline and, separately, the CEV of the population alive in 2030 when the remainder is revealed, against the half-decline path. Newborns: mean CEV of the cohorts born 2023–2122.

Energy Agency's net-zero scenario, in which the oil price falls to 25 dollars per barrel by 2050, about 0.3 of its 2023 level; International Energy Agency, 2024), the year by which the decline is completed (2040, 2050 or 2060, with the carbon charge on the same timing), and the credibility of the announcement (the 2022 announcement covers half of the decline and the remainder is revealed in 2030).

The fossil-price mechanism increases with the terminal fall, but less than proportionally: −1.23 points at a terminal level of 0.70, −1.65 at 0.56, −1.98 at 0.40 and −2.12 at 0.30, because fossil output and the export base shrink as the price falls, so each further dollar of decline is applied to a smaller base. The two channels are equal at a terminal level of about 0.93, a decline of about 7 percent. This break-even point is extrapolated linearly between no decline and the smallest decline I solve (a terminal level of 0.70), so it is an approximation; but the price mechanism exceeds the tax mechanism by a factor of four even at that smallest decline, and any scenario of global decarbonization implies a decline several times larger than 7 percent (the baseline's is 44 percent). A faster decline (completed by 2040) raises the mechanism to −2.23 points and a slower one (2060) lowers it to −1.27, because the rents lost earlier are discounted less.

When only half of the decline is announced in 2022, the 2022 population loses −1.27 percent. When the remainder is revealed in 2030, the households alive then lose a further −0.86 percent against the half-decline path, −1.61 for those aged 20–34 and −0.40 for those over 65. Because the surprise arrives before most of the price fall, little of the adjustment has been made on the wrong path. A higher adjustment cost of fossil capital, which strands capital in the sector, leaves the mean loss under the announced decline almost unchanged (−2.002 against −1.996 percent) but raises the cost of the 2030 surprise from −0.86 to −0.96 percent.